Statistical Signal Processing
Lecture 06:
Extended (linearizing) Kalman filter for non-linear systems

Literature and Problems: See course schedule

Agenda (preliminary):

Linear signal and obs model for KF

- **non-linear models ?**
- **non-gaussian / multimodal process / obs noise ?**
- **EKF linearizes around current state estimate**

Non-linear process model: $\mathbf{x_k} = \mathbf{f}(\mathbf{x_{k-1}}, \mathbf{u_{k-1}}, \mathbf{w_{k-1}})$
Non-linear obs model: $\mathbf{z_k} = \mathbf{h}(\mathbf{x_k}, \mathbf{v_k})$

Linearization of non-linear function, $\mathbf{y} = \mathbf{f}(\mathbf{x})$, **around** $(\mathbf{x_0}, \mathbf{y_0})$ **in general**

EKF linearizes around the currently predicted state

- $\tilde{\mathbf{x}}_\mathbf{k} = \hat{\mathbf{x}}_\mathbf{k}^- = \mathbf{f}(\hat{\mathbf{x}}_{\mathbf{k-1}}, \mathbf{u_{k-1}}, \mathbf{0})$
- $\tilde{\mathbf{z}}_\mathbf{k} = \hat{\mathbf{z}}_\mathbf{k}^- = \mathbf{h}(\hat{\mathbf{x}}_\mathbf{k}^-, \mathbf{0})$

Linearized process model around predicted state

- $(\mathbf{x_k} - \hat{\mathbf{x}}_\mathbf{k}^-) = \mathbf{A_k}(\mathbf{x_{k-1}} - \hat{\mathbf{x}}_{\mathbf{k-1}}) + \mathbf{W_k}(\mathbf{w_{k-1}} - \mathbf{0})$
- **Jacobi matrices** $\mathbf{A_k}, \mathbf{W_k}$

Linearized observation model around predicted state

- $(\mathbf{z_k} - \hat{\mathbf{z}}_\mathbf{k}^-) = \mathbf{H_k}(\mathbf{x_k} - \hat{\mathbf{x}}_\mathbf{k}^-) + \mathbf{V_k}(\mathbf{v_k} - \mathbf{0})$
- **Jacobi matrices** $\mathbf{H_k}, \mathbf{V_k}$

EKF predict step equations: $\hat{\mathbf{x}}_\mathbf{k}^-,\ \hat{\mathbf{z}}_\mathbf{k}^-,\ \mathbf{P}_\mathbf{k}^-$
EKF update step equations: $\hat{\mathbf{x}}_\mathbf{k}\ , \mathbf{K_k},\ \mathbf{P_k}$
Extended Kalman filter algorithm block diagram including initialization

Limitations of EKF

- **EKF is "ad hoc" filter and not "optimal", but best linear filter**
- **EKF best for weakly non-linear systems and close to gaussian noises**
- **EKF can diverge**

Demos

- **EKF vs. KF for linear and non-linear systems**